

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [115, 114, 119, 113, 117, 111, 107, 115, 115, 122]

Tap amplitudes (pixels): [185.88437266214714, 179.25400971805345, 167.88686666919483, 174.40183485273312, 173.60011520733505, 162.8649747490233, 170.07351351694948, 160.4306703844374, 173.92527130926086, 176.11927776367924, 177.50492950901392]

GPT-5

MDS-UPDRS Item 3.4 Finger Tapping – Feature-Based Assessment

What the data show

- Instructional target: 10 taps
- Detected taps: 11 (likely includes an extra start/end peak; we evaluate the first 10 intervals/taps)
- Tap periods (frames, n=10 intervals): [115, 114, 119, 113, 117, 111, 107, 115, 115, 122]
 - Central tendency: mean $\approx$ 114.8 frames; median $\approx$ 115
 - Variability: SD $\approx$ 4.1 frames (low); range 111–122 (spread = 11 frames, $\sim$ 9.6% of mean)
 - Rhythm: steady, no obvious long arrests/freezes; no abrupt outliers
 - Trend: slight shortening toward taps 6–7 (111 $\rightarrow$ 107) then mild lengthening to 122 by the last interval; overall quite regular
- Tap amplitudes (pixels, n=11 peaks): [185.9, 179.3, 167.9, 174.4, 173.6, 162.9, 170.1, 160.4, 173.9, 176.1, 177.5]
 - Pattern: clear early decline from tap 1 $\rightarrow$ 3 (-9.7% then -6.4% vs prior), intermittent partial recovery thereafter
 - Minimum around tap 8 ($\sim$ 160.4, -13.7% vs tap 1), then rebound to $\sim$ 177–178 by the end

- Net from first to last: -4.5% (moderate early decrement with partial compensation)
- Decrement timing: begins within the first few taps (after tap 1), not just near the end and more than a mid-sequence-only dip

Mapping to scoring criteria

- Interruptions/hesitations: Period series is regular with small variability; no evidence of 3–5 interruptions or >5 interruptions or a clear freeze. This supports 0–1 rather than 2–3 on interruption criterion.
- Speed: Low variability and periods clustered near the mean suggest no clear mild/moderate slowing relative to the subject's own sequence; we don't have absolute time calibration or normative frames → seconds mapping, so we infer speed from consistency. This leans away from 2–3 for slowing.
- Amplitude behavior:
 - Not “near the end only” (which would favor 1).
 - Not strictly “midway through” only (which would favor 2).
 - There is a notable decrement starting after the 1st tap (down to taps 2–3 and again to 6–8), which aligns with the “3 (Moderate): amplitude decrements starting after the 1st tap.”
 - Despite later partial recovery, the early decrement is present and fairly pronounced (~10–14% from first to local minima), satisfying the 3-point amplitude criterion.

Given that the amplitude criterion alone is sufficient to set the score at the highest applicable level among criteria, and interruptions/slowing do not contradict this, the best fit is: